Quantities and unit system

→ To measure a quantity is to compare it to another of the same nature taken as a unit

1) Definitions

1) Quantity: is a measurable or estimable dimension

2) Unit: is a reference quantity used as a basis for comparison with other quantities of the same kind

3) Value: This represents how many times the standard (unit) must be repeated to obtain the desired measurement

Quantity ← m = 35 Kg ⇒ Unit
↓
Value

2) Categories of quantities:

1) Extensive quantities: These are additive (the total quantities of two objects is the sum of their individual quantities)
المقادير الكبيرة — مضافة

2) Intensive quantities: These are not additive
المقادير المكثفة

3) Unit systems:

1) Système MKSA:

The fundamental units are:

→ meter (m): Unit of distance
→ Kilogram (Kg): Unit of mass
→ Second (s): Unit of time
→ ampere (A): Unit of electric curren
شدة التيار الكهربائي

2) CGS system:

The fundamental units are:

→ Centimeter (cm): $1cm = 10^{-2} m$
→ gram (g): $1g = 10^{-3} kg$
→ Second (s): Unit of Time

3) International system of units (SI

⇒ The modern system with seven fundamental units:

Quantity	Symbol	Unit	Symb
Length	L	Meter	m
Mass	M	Kilogram	Kg
Time	T	Second	S
Electric current intensity	I	Amper	A
Temperature	Q	Kelvin	K
Amount of substance كمية المادة	N or N	Mole	mo
Luminous Intensity شدة الإضاءة	J	Candela	c

⇒ Added:

Quantity	Symbol	Unit	Sy
Plan Angle	α	Radian	ra
Solid Angle			

⇒ Some derived units:

Quantity	Symbol	Unit	Symbol
Frequency التردد	ν	Hertz	s^{-1}
Force قوة	F	Newton	$m.kg.s^{-2}$
Pressure ضغط	P	Pascal ($Pa = N.m^{-2}$)	$m^{-1}.kg.s^{-2}$
Electric potential difference فرق الجهد	V	Volt ($V = W.A^{-1}$)	$M^2.kg.s^{-3}.A$

3) Homogeneity of relations:

1) Dimensional relation
العلاقات البعدية

⇒ A relation in physics must be homogeneous in terms of units, meaning that the units in the first member of the relation must have the same meaning as those in the second member

⇒ A relation that is not homogeneous in terms of units is false. The systems of units may be different in the two members of the relation, but the relation must be homogeneous

⇒ Any non-homogeneous result is necessarily false, however, a homogeneous result is not necessarily correct

⇐ العلاقة في الفيزياء لازم تكون متجانسة
من حيث الوحدات (system of units) يعني
الوحدة في ط1 لها نفس المعنى في ط2
⇐ يمكن أن يكون system of units مختلف في
الطرفين لكن يجب أن تكون متجانسة
⇐ أي علاقة غير متجانسة بالضرورة خاطئة، لكن
العلاقة المتجانسة ليست بالضرورة صحيحة

2) Dimensionless relation:

⇒ A relation may not involve units

⇒ It is the ratio between two quantities of the same kind

⇒ A constant value within the relation may carry units:

⇒ for the relation to be homogeneous, the constant ... must be expressed in:

2) ⇐ يمكن أن تكون هنا علاقة بدون وحدة
⇐ يمكن أن تكون كذلك مقادير لا يحتوون
وحدة لأن الكميات (الكميتين لهم نفس
في نفس الطبيعة ولهم نفس القانون
3) ⇐ يمكن حتى Constant في العلاقة يبقى
وحدة ونخرجها في العلاقة

4) The coherence of relations:

⇒ A relation is coherent if it doe
not involve a constant (other tha

coherent ≠ incoherent

لاقة التي لا تحتوي على ثابت غير 1

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